

Saliency-Aware Quantized Imitation Learning for Efficient Robotic Control

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Abstract

Deep neural network (DNN)-based policy models, such as vision-language-action (VLA) models, excel at automating complex decision-making from multi-modal inputs. However, scaling these models greatly increases computational overhead, complicating deployment in resource-constrained settings like robot manipulation and autonomous driving. To address this, we propose Saliency-Aware Quantized Imitation Learning (SQIL), which combines quantization-aware training with a selective loss-weighting strategy for mission-critical states. By identifying these states via saliency scores and emphasizing them in the training loss, SQIL preserves decision fidelity under low-bit precision. We validate SQIL’s generalization capability across extensive simulation benchmarks with environment variations, real-world tasks, and cross-domain tasks (self-driving, physics simulation), consistently recovering full-precision performance. Notably, a 4-bit weight-quantized VLA model for robotic manipulation achieves up to $2.5\times$ speedup and $2.5\times$ energy savings on an edge GPU with minimal accuracy loss. These results underline SQIL’s potential for efficiently deploying large IL-based policy models on resource-limited devices.

1. Introduction

In recent years, deep neural network (DNN)-based policy models have significantly advanced robot manipulation and autonomous driving [3, 4, 10, 26, 32, 54], primarily by surpassing traditional search-based methods through imitation learning (IL) from expert data. This success has led to growing interest in adopting foundation models for large-scale IL, aiming to improve generalization and robustness beyond limited data, facilitating policy transfer across different robot embodiments, tasks, or environments [3, 4]. Vision-

Language Action (VLA) models [3, 4, 10, 26] extend pre-trained vision-language models to robotics via next-token prediction, integrating visual and textual information for enhanced manipulation capabilities. Despite their potential as foundation models supporting cross-embodiment transfer, IL-based VLA models often suffer from slow inference speeds, high computational costs, and substantial memory usage [49], making deployment on resource-constrained, battery-powered robots challenging.

Quantization techniques reduce DNN inference costs by converting full-precision (FP)¹ weights and activations to lower precision [6, 11, 15, 33], and extensive research focuses on mitigating the resulting numerical errors to preserve accuracy. Post-training quantization (PTQ) adjusts weights and activations to reduce quantization loss, whereas quantization-aware training (QAT) incorporates quantization directly in training to enhance robustness. These strategies are well studied in computer vision [30, 37, 51, 57] and natural language processing [14, 33, 50], but their effects on IL-based applications, such as robot manipulation or autonomous driving, remain underexplored.

In this work, we investigate how quantization errors affect IL policy decisions and note two key observations: 1) Quantization errors generally have minor impact across most timesteps, producing small action discrepancies (e.g., t_1 in Fig.1). 2) Certain critical states, however, experience large deviations in actions due to quantization errors (Fig.1- t_2 : PTQ, t_3 : QAT), ultimately causing mission failures (Fig.1- t_4). This differs from typical long-sequence prediction, where errors accumulate at each timestep to deviate from the FP policy’s action distribution; here, the quantized policy must specifically address a small number of mission-critical states.

To address this, we present saliency-aware quantized imitation learning (SQIL), a simple yet effective method for IL-based robotic control. SQIL integrates quantization into

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¹We use BFloat16 [24] and Float32 as baseline FP for OpenVLA and CILRS, unless noted otherwise.

Task: **put the black bowl in the bottom drawer of the cabinet and close it**

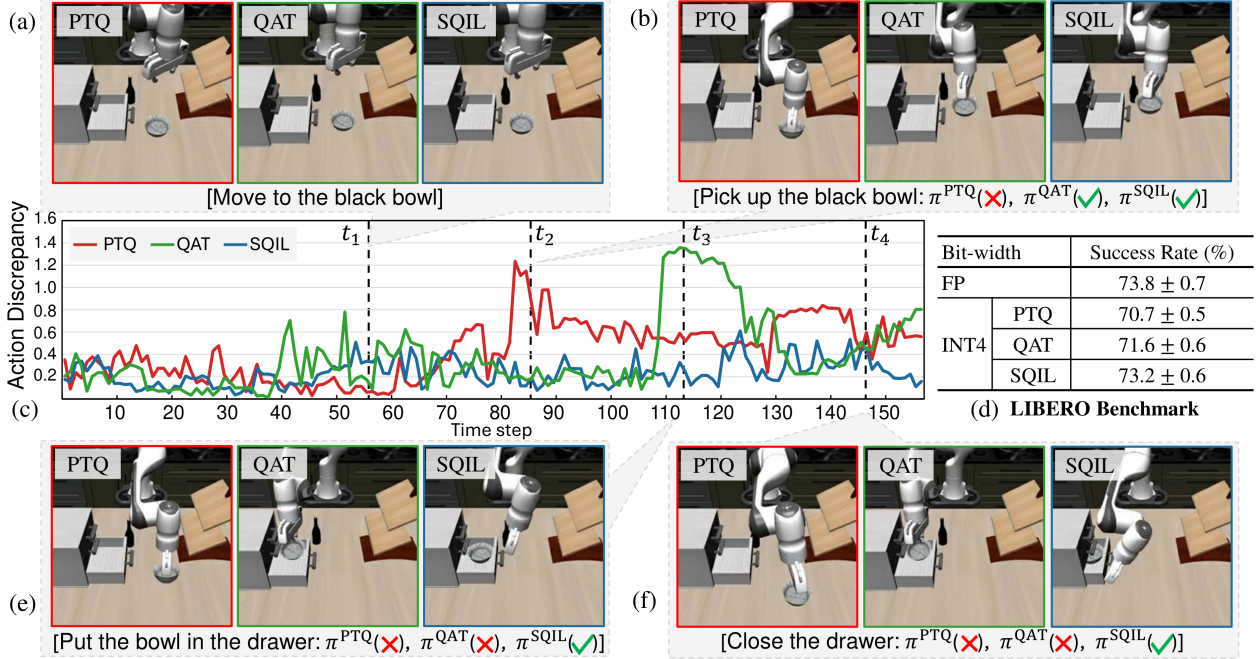


Figure 1. Comparison of action discrepancy (L2-norm) between the quantized and full-precision (FP) policies in *OpenVLA* using the PTQ, QAT, and SQIL, and evaluation of the average success rate on the LIBERO benchmark. For fair comparison, episode timelines are segmented into four sub-tasks and realigned based on successful transitions.

fine-tuning, increasing robustness against quantization errors. Unlike traditional QAT for supervised learning, SQIL incorporates quantization-robust action distillation (QRD): 1) identifying mission-critical states via saliency scores, and 2) applying an importance-weighted loss to emphasize correct actions at these states. Consequently, the quantized policy more closely follows the FP policy’s decisions for crucial states. As seen in Fig.1, SQIL suppresses action discrepancies at critical timesteps (t_2 , t_3), leading to mission success and fully recovering the FP’s success rate (Fig.1(d)).

We validate SQIL’s generalization capability across extensive simulation benchmarks with environment variations, real-world tasks, and cross-domain tasks (self-driving, physics simulation), consistently recovering full-precision performance. For robot manipulation, evaluations of *OpenVLA*[26] on the LIBERO benchmark[34] show that our 4-bit weight-quantized models achieve success rates comparable to the FP policy, delivering a 2.5 $\times$ speedup and 2.5 $\times$ energy savings on edge GPUs. In autonomous driving, tests of *CILRS*[54] on the NoCrash benchmark[8] confirm that our 4-bit weight- and activation-quantized policy maintains the FP policy’s performance, realizing a 3.7 $\times$ speedup and 3.1 $\times$ energy savings on low-end GPUs. These findings mark the first successful demonstration of recovering and efficiently deploying quantized IL-based policies.

2. Related Works

2.1. DNN-based Policy for Imitation Learning

DNN-based policy models trained via imitation learning (IL) have propelled significant progress in physics simulation, autonomous driving, and robotic manipulation. In MuJoCo [46], for example, DNN models control robots in continuous joint spaces by processing state-vector inputs, often trained on benchmarks such as *D4RL*[16]. Autonomous driving models like *CILRS*[54] learn to predict acceleration and steering from camera images and measurement vectors, focusing on safe driving. In robotic manipulation, vision-language-action models such as RT-2 [4] and *OpenVLA*[26] use LLMs to fuse visual and textual inputs for multi-task tasks; *OpenVLA* specifically generates actions (x, y, z, roll, pitch, yaw, grasp) auto-regressively from image observations and language instructions. These IL-based models learn by imitating expert behaviors from pre-collected datasets. For instance, *CILRS* is trained on 80 demonstration, while *OpenVLA* is pretrained on 970k robot episodes and then fine-tuned on 1,700 LIBERO simulation episodes [34] with LoRA [21] for training efficiency.

As IL applications grow more complex, DNN-based policy models have dramatically increased in size. Fig.2(c-e) shows representative IL models for physics simulation (*D4RL*), autonomous driving (*CILRS*), and vision-language-action (*OpenVLA*), with model sizes in Fig.2(b).

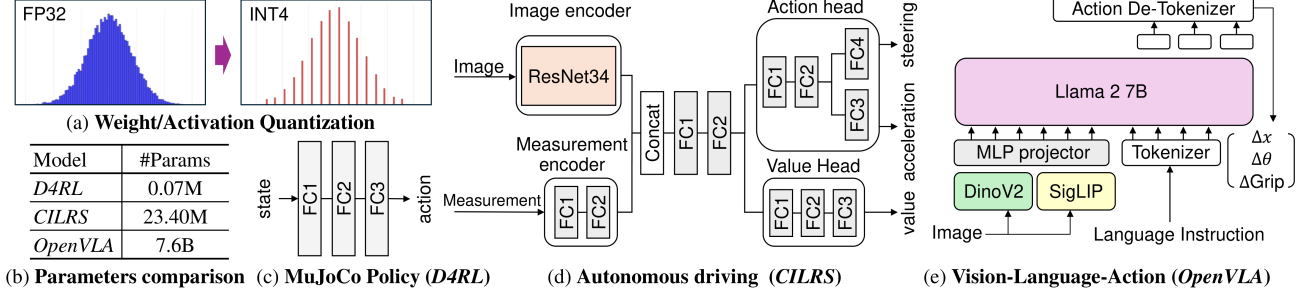


Figure 2. Comparison of the structure, number of parameters of DNN-based policy models.

D4RL relies on a three-layer fully connected network, while *CILRS* combines a ResNet-based image encoder with fully connected layers for steering control. By contrast, *OpenVLA* merges the Llama2-7B language model [47] with visual features from DINOv2 [40] and SigLIP [52], creating a versatile foundation for robotic manipulation. This expansion has increased model sizes by about 100,000 times (from 0.07M parameters in D4RL to 7.6B in *OpenVLA*). Since scaling model size often boosts capability [25], efficient inference methods are needed to reduce compute and memory loads on resource-constrained devices.

2.2. Quantization for Efficient Policy Inference

Quantization reduces neural networks’ computational and memory requirements by lowering the bit-precision of weights and/or activations, typically applied to the input operands of general matrix multiplication (GEMM). In compute-intensive convolutional models, both weights and activations are quantized [7, 19, 43], increasing operations per second [35] and boosting energy efficiency [20]. Large language models (LLMs), on the other hand, often adopt weight-only quantization [15, 33] to reduce memory-access bottlenecks.

Numerical errors can degrade accuracy since quantization maps continuous values to discrete states (Fig.2(a)). Post-training quantization (PTQ) employs data manipulations [1, 28, 44]—for instance, AWQ [33] shapes weight distributions for weight-only quantization in LLMs—though it does not inherently improve resilience to quantization errors². In contrast, quantization-aware training (QAT) incorporates quantization effects during training to maintain accuracy [6, 11, 22, 56], sometimes combined with knowledge distillation (KD) [36, 55, 58] for further error compensation. For LLMs, where tuning all parameters is expensive, LoRA [9, 18, 31] refines only small adapters while keeping base parameters in reduced precision.

Despite rising demand for efficient policy inference, most quantization approaches remain tailored to reinforcement learning (RL), which relies on direct environment in-

teraction—unavailable in IL—to mitigate quantization errors. Early work like low-precision policy distillation (LPPD)[36] used aggressive 1-bit QAT for classical policy models. More recent RL studies[13, 53] have quantized lightweight policies, leveraging environment interaction to compensate for errors; structured pruning [42] similarly reduces model size without hurting decision quality. However, these methods do not directly apply to IL, which lacks environment feedback. OpenVLA [26] explored QAT with LoRA for 4-bit quantization but saw mixed success. Our investigation shows that conventional QAT suffers noticeable accuracy loss on OpenVLA, underscoring the need for more advanced quantization-error mitigation methods for IL.

3. Background and Challenges

3.1. Imitation Learning

Imitation Learning (IL) enables an agent to learn policies directly from expert demonstrations, which is particularly useful when reward functions are hard to define (e.g., robotics, autonomous driving). Let S be the set of states, and A the set of actions. The expert dataset $D_E = \{\tau_1, \dots, \tau_N\}$ contains N expert demonstrations, each τ_i being a sequence of state-action pairs of length T_i , denoted as $\tau_i = \{(s_1, a_1), \dots, (s_{T_i}, a_{T_i})\}$, where $s_t \in S$ and $a_t \in A$.

The goal of IL is to learn a policy $\pi_\theta^{\text{FP}} : S \rightarrow A$ that replicates expert behavior. A common approach is *behavior cloning* [41], which maximizes the likelihood of expert state-action pairs via:

$$\mathcal{L}^{\text{IL}}(\theta) = -\mathbb{E}_{\tau_i \sim \mathcal{D}_E} \left[\frac{1}{|\tau_i|} \sum_{(s_t, a_t) \in \tau_i} \log \pi_\theta^{\text{FP}}(a_t | s_t) \right]. \quad (1)$$

Minimizing this loss guides π_θ^{FP} to imitate the expert actions a_t for corresponding states s_t .

3.2. Policy Quantization

As IL extends to more complex robotic control tasks, DNN-based policy models can reach billions of parameters, demanding substantial computation and memory. To address this, quantization is applied to weights and/or activa-

²We use AWQ [33] and RTN [23] as our baseline PTQ methods for OpenVLA and others, respectively, unless otherwise noted.

Task: **pick up the black bowl** in the top drawer of the wooden cabinet and **place it on the plate**

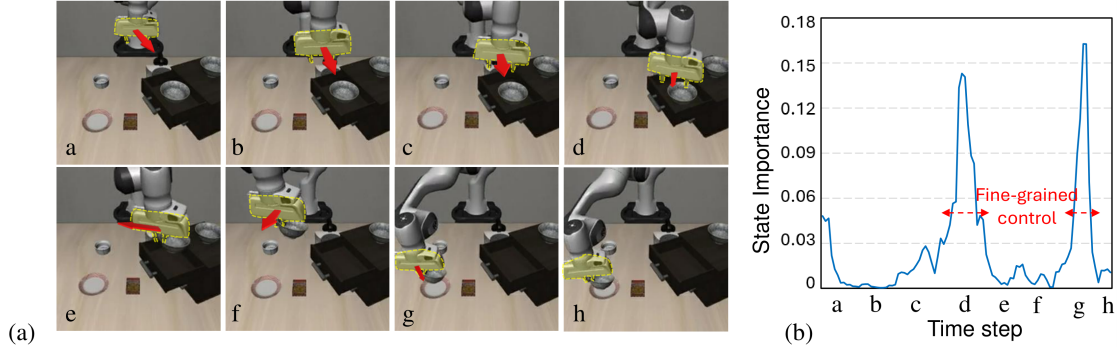


Figure 3. Comparison of timestep-wise state importance in the *OpenVLA* robot manipulation task.

tions (e.g., weight and activation quantization for compute-bounded CNNs, weight-only quantization for memory-bounded LLMs). We convert full-precision data w^{FP} to b -bit precision using:

$$w^Q = \text{Clip} \left(\left\lfloor \frac{w^{FP}}{\gamma} \right\rfloor, -2^{b-1}, 2^{b-1} - 1 \right), \quad (2)$$

where γ is the quantization scale. The resulting quantized policy model π_θ^Q reduces computation and memory use but may produce action discrepancies due to quantization errors. QAT incorporates quantization into IL with:

$$\mathcal{L}^{\text{QAT}}(\theta) = -\mathbb{E}_{\tau_i \sim \mathcal{D}_E} \left[\frac{1}{|\tau_i|} \sum_{(s_t, a_t) \in \tau_i} \log \pi_\theta^Q(a_t | s_t) \right]. \quad (3)$$

This allows π^Q to approximate expert behavior similar to π^{FP} . The same expert dataset $\mathcal{D}_E$ is reused to update π_θ^Q 's weights, and the same hyperparameters (learning rate, etc.) can typically be used without hassle.

3.3. Challenges

Although PTQ and QAT mitigate some quantization errors, they still face noticeable performance drops, especially in success rate. As shown in Fig. 1, comparing the action distributions of a full-precision VLA policy with INT4 PTQ and QAT reveals two main points: 1) quantization errors usually result in minor discrepancies across most timesteps (t_1); and 2) at certain *mission-critical* states, such as ‘‘Pick-up the black bowl’’ (t_2) or ‘‘Put the bowl in the drawer’’ (t_3), quantization causes large action deviations (π^{PTQ} at t_2 and π^{QAT} at t_3), leading to mission failures (t_4)³.

These observations differ from *distribution shift* [29], which arises when encountering unseen or out-of-distribution states. Here, the deviation stems from model quantization rather than dataset mismatch (QAT still uses the same expert dataset), and only a few critical states fail

instead of the entire trajectory. These observations suggest quantized policies must specifically address those critical states. Developing methods that minimize quantization errors at these points is crucial for reliable robotic control performance in dynamic, complex environments.

4. Method

Motivated by the previous observations, we propose enabling the quantized policy to learn compensations for quantization errors in mission-critical states. The proposed method, saliency-aware quantized imitation learning (SQIL), consists of two key components: 1) a saliency-based state-importance score (SIS) to identify mission-critical states without supervision, and 2) a quantization-robust action distillation (QRD) method that selectively emphasizes learning from the SIS-identified mission-critical states.

4.1. Saliency-based State-Importance Score

From Fig. 1, we see that mission-critical states depend on both the task and the environment, making it non-trivial to identify which states need extra attention during imitation learning. Therefore, we quantify the state importance score (SIS) to identify mission-critical states. Motivated by [17], we maintain that states showing large action discrepancies under visual perturbation are mission-critical, as these discrepancies reveal the crucial visual regions for successful decision-making. We measure the perturbation-based action saliency at each position of a state as follows:

$$S_\pi(s_t, k) = \frac{1}{2} \|\pi(s_t) - \pi(\phi(s_t, k))\|^2, \quad (4)$$

where $\phi(s_t, k)$ applies a perturbation (e.g. a Gaussian blur at position k). Higher $S_\pi(s_t, k)$ indicates that local modifications at k have a greater impact on the policy’s output

We then define SIS as the average saliency across all positions k :

$$\text{SIS}_\pi^{s_t} = \mathbb{E}_k [S_{\pi^{FP}}(s_t, k)]. \quad (5)$$

³A similar effect is observed in self-driving tasks.

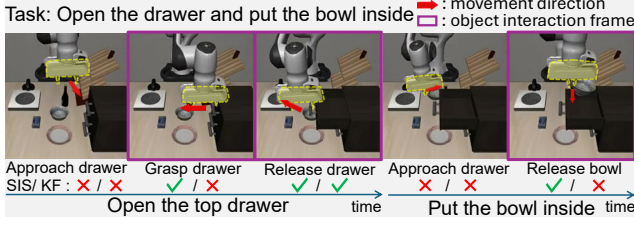


Figure 4. Comparison of SIS and keyframe (KF) on trajectory.

LIBERO Benchmark	QAT	SQIL (KF)	SQIL (SIS)
Avg. Success Rate (%)	71.6 $\pm$ 0.5	72.6 $\pm$ 0.7	73.2 $\pm$ 0.6

Table 1. Saliency metric comparison: Keyframe (KF) and SIS.

A higher SIS suggests that the state has more perturbation-sensitive positions, often found in mission-critical scenarios. As shown in Fig. 3, SIS effectively highlights the states at the “Pick-Up” (timestep d) and “Drop” (timestep g) actions, highlighting its ability to detect moments requiring fine-grained control. Compared to vision-language-based key-frame (KF) detection [27], which often activates only at coarse sub-task boundaries (e.g., “drawer open”, “object placed”), SIS responds to subtle yet decisive robot–environment interactions—such as initial grasp or release—by measuring how small perturbations alter the policy’s output (Fig.4). Quantitatively, replacing SIS with KF in SQIL reduces success by 1.1% on LIBERO manipulation tasks (Table 1), confirming that policy-driven saliency is essential to robust performance under quantization.

4.2. Quantization-Robust Action Distillation

We selectively focus on mission-critical states using the state importance score (SIS). To achieve this, we propose quantization-robust action distillation (QRD), which leverages the FP policy and demonstration data to reduce quantization errors by distilling the FP policy’s action distributions into the quantized model. Its loss function measures the discrepancy between the quantized and FP policies’ action distributions:

$$\mathcal{L}^{\text{QRD}}(\theta) = \alpha_t \cdot \mathbb{E}_{\tau_i \sim \mathcal{D}_E} \left[\frac{1}{|\tau_i|} \sum_{s_t \in \tau_i} D(\pi^Q(s_t), \pi^{\text{FP}}(s_t)) \right], \quad (6)$$

where $\alpha_t = \beta$ if $SIS_{\pi^{\text{FP}}}^{s_t} > T$, and $\alpha_t = 1$ otherwise; D is a discrepancy metric (e.g., L2-norm), and $\pi(s_t)$ is the probability distribution of all possible actions for state s_t . The hyperparameter $\beta (> 1)$ applies extra weight to high-importance states, and T is a threshold that selects the top $p=20\%$ of SIS values. Unlike conventional KD methods, QRD employs the weighting α_t to amplify the loss for states identified by SIS (i.e., $SIS_{\pi^{\text{FP}}}^{s_t} > T$). Training convergence is not sensitive to the hyperparameters $D(), \beta, T$, and we use the same values (details in Supplementary Sec. ??) for

Algorithm 1 Saliency-Aware Quantized Imitation Learning

Input: Pre-trained policy π^{FP} , quantized policy π_θ^Q , Expert dataset $\mathcal{D}_E$, Number of epochs N
Output: Quantized policy π_θ^Q
Initialize π_θ^Q from π^{FP}
Calculate $SIS_{\pi^{\text{FP}}}$ with π^{FP} , $\mathcal{D}_E$ using Eq. (5)
for epoch = 1 to N **do**
 for each trajectory τ_i in $\mathcal{D}_E$ **do**
 for each state-action pair (s_t, a_t) in τ_i **do**
 Calculate $\mathcal{L}^{\text{QAT}}$ using Eq. (3)
 Calculate $\mathcal{L}^{\text{QRD}}$ with $SIS_{\pi^{\text{FP}}}$ using Eq. (6)
 $\mathcal{L}^{\text{SQIL}} = \mathcal{L}^{\text{QAT}} + \mathcal{L}^{\text{QRD}}$ (Eq. (7))
 Compute $\frac{\partial \mathcal{L}^{\text{SQIL}}}{\partial \theta} \approx \frac{\partial \mathcal{L}^{\text{SQIL}}}{\partial \theta_Q}$
 Update π_θ^Q with $\frac{\partial \mathcal{L}^{\text{SQIL}}}{\partial \theta}$
 end for
 end for
end for
return Quantized policy π_θ^Q

all the experiments across robot control, self-driving, and physics simulation.

4.3. Saliency-Aware Quantized Imitation Learning

To enhance conventional QAT for mission-critical states, we introduce saliency-aware quantized imitation learning (SQIL), which selectively strengthens IL’s weight adjustment for salient states. SQIL’s loss function combines QAT and QRD:

$$\mathcal{L}^{\text{SQIL}}(\theta) = \mathcal{L}^{\text{QAT}}(\theta) + \mathcal{L}^{\text{QRD}}(\theta), \quad (7)$$

This combined approach effectively reduces quantization errors, enabling the quantized policy to generalize comparably to the FP policy. $\mathcal{L}^{\text{QAT}}$ maximizes the quantized policy’s log-likelihood of expert actions, while $\mathcal{L}^{\text{QRD}}$ aligns the quantized policy’s action distribution with the FP policy. By applying selective weighting α_t , QRD emphasizes states requiring precise control instead of uniformly minimizing the discrepancy across all states.

Algorithm 1 outlines the overall SQIL procedure. We reuse the same expert dataset and training hyperparameters, and SIS can be precomputed once for $\mathcal{D}_E$ and π^{FP} . Consequently, SQIL offers a turnkey solution for efficiently quantizing an FP policy using an existing expert dataset. As shown in Fig. 1, SQIL suppresses action discrepancies at mission-critical timesteps (t_2, t_3), leading to mission success and fully recovering the FP’s success rate (Fig. 1(d)).

4.4. Analysis

In this section, we quantitatively answer two questions about SQIL’s efficacy: 1) Does it restore the disrupted saliency of the quantized policy? 2) Do QAT and QRD

TASK : **pick up the black bowl** between the plate and ramekin and **place it on the plate** (red dashed circle) : attended nonessential region

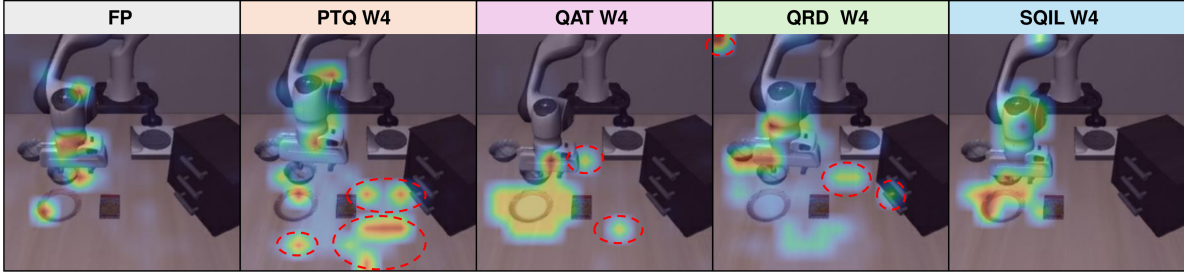


Figure 5. Comparison of attention visualization in **important states** (pick up the black bowl) for tasks successfully completed on the LIBERO-Spatial benchmark. Additional examples are provided in the *supplementary materials*.

Method	Average Saliency Divergence↓			
	LIBERO -Spatial	LIBERO -Object	LIBERO -Goal	LIBERO -Long
PTQ	0.0821	0.0268	0.0612	0.0640
SQIL	0.0559	0.0198	0.0458	0.0521

Table 2. Comparison of AttDiv with *OpenVLA* on the LIBERO.

within SQIL work synergistically to recover the action density of the quantized policy?

Saliency Visualization. To qualitatively assess how quantization affects IL policy behavior and how SQIL addresses this issue, we visualize the saliency map using Eq. 4. Fig. 5 shows a robot executing “Pick-up the black bowl” under different policies. The FP policy exhibits high saliency on objects of interest (e.g., the robot arm, the target bowl, and the destination plate). In contrast, the PTQ policy focuses on irrelevant regions, indicating that quantization errors misidentify salient areas. Although QAT and QRD alone attempt to restore expert or FP actions, applying loss with uniform importance across all states limits their ability to fix the saliency. In contrast, SQIL’s saliency map closely mirrors the FP policy, indicating that it effectively recovers the quantized policy’s desired focus. We confirm this quantitatively in Table 2, which shows SQIL achieving lower average divergence from the FP saliency map than PTQ.

Action Distribution Comparison. To evaluate the synergy of QAT and QRD, we compare the action distributions of the FP policy and policies quantized by PTQ, QAT, QRD, and SQIL in the self-driving model (CILRS). Fig. 6 depicts each policy’s distribution, with circle and diamond markers indicating the maximum-likelihood actions of the FP and quantized policies, respectively. PTQ introduces large quantization errors, causing serious divergence from FP. Although QAT aligns the maximum likelihood to expert actions, it can generate overly sharp peaks that distort the FP distribution. QRD aligns the quantized distribution with the FP policy’s shape but may overlook expert actions when the FP distribution is broad. By combining both, SQIL preserves the FP policy’s overall behavior while emphasizing high-quality expert actions, resulting in better decisions and improved performance.

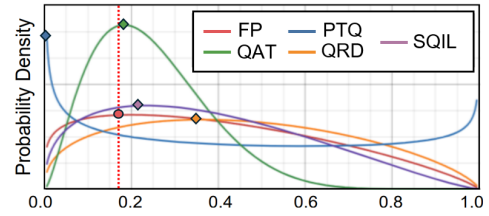


Figure 6. Comparison of action distributions for various quantization methods using *CILRS* (W4A4) on NoCrash-dense at state s_j . The red circle denotes action a_j from the demonstration dataset $\mathcal{D}_E$, and the diamond shape marks the highest probability point for each distribution.

5. Experiments

5.1. Experimental Settings

We conducted experiments on the quantization of DNN-based policy models across various domains, including robot manipulation, autonomous driving, and classical physics simulation. Detailed experimental settings for each task are provided in the *supplementary materials*.

Robot Manipulation: We utilized *OpenVLA* [26] with weight quantization through AWQ [33] and QuaRot [12] methods. Evaluation were made using the the LIBERO [34] benchmark in simulated environments, spanning tasks like LIBERO-Spatial, LIBERO-Object, LIBERO-Goal, and LIBERO-Long. Each task was repeated in three rounds of evaluations. We fine-tuned a minimal set of trainable parameters (110M parameters) using SQIL and QLoRA (r=32) and tested the model in two real-world scenarios: a replication of BridgeDataV2 [48] and a tabletop setup that enables broader manipulation behaviors. A dataset $\mathcal{D}_E$ of 80 episodes was collected and used to further refine *OpenVLA* with SQIL and QLoRA for deployment on the UR5 robot.

Autonomous Driving: We employed *CILRS* [54] and enhanced latency by applying both weight and activation quantizations at 4-bit precision, using RTN [23] for PTQ and LSQ [11] for SQIL. The models were evaluated on the NoCrash-dense benchmark [8], which tests performance across different towns and weather conditions.

Classical Physics Simulation: Fine-tuning and assess-

Method	Quantizer (INT4)	Success Rate % $\uparrow$			
		LIBERO-Spatial	LIBERO-Object	LIBERO-Goal	LIBERO-Long
FP	-	84.0 $\pm$ 0.9	83.9 $\pm$ 0.3	76.6 $\pm$ 0.6	50.7 $\pm$ 1.2
PTQ	AWQ	80.1 $\pm$ 0.5	81.3 $\pm$ 0.4	74.3 $\pm$ 0.6	47.2 $\pm$ 0.6
QAT	AWQ	80.9 $\pm$ 0.7	82.4 $\pm$ 0.4	75.7 $\pm$ 0.5	47.3 $\pm$ 0.3
SQIL	AWQ	83.9 $\pm$ 0.5	83.5 $\pm$ 0.5	76.3 $\pm$ 0.4	49.2 $\pm$ 1.0
PTQ	QuaRot	81.2 $\pm$ 1.0	81.7 $\pm$ 0.6	74.8 $\pm$ 0.4	47.6 $\pm$ 0.9
QAT	QuaRot	81.8 $\pm$ 0.9	82.8 $\pm$ 0.5	75.2 $\pm$ 0.6	48.0 $\pm$ 0.8
SQIL	QuaRot	83.8 $\pm$ 0.8	83.7 $\pm$ 0.5	76.3 $\pm$ 0.5	49.4 $\pm$ 0.9

Table 3. Comparison of success rate across various INT4 quantization with *OpenVLA* and scenarios in the LIBERO benchmark.

Language Instruction Generalization				Illumination Generalization		
Task	#Trials	# Successes $\uparrow$		Light Intensity	Success Rate(%) $\uparrow$	
		FP	SQIL W4		FP	SQIL W4
Pick up the black bowl on the wooden cabinet and place it on the plate	50	34	29	100%	50.7	49.2
Grab the black bowl from the wooden cabinet and set it on the plate	50	25	28	80%	50.5	49.1
Remove the black bowl from the wooden cabinet and place it on the plate	50	31	34	60%	49.5	48.8
Lift the black bowl from the wooden cabinet and set it on the plate	50	28	27	—	—	—

Table 4. Generalization comparisons of *OpenVLA*: Language instruction generalization in LIBERO-Spatial (left) and illumination generalization in LIBERO-Long (right).

LIBERO Benchmark	FP	PTQ	QAT	SQIL
Model Size	3.3B	1.1B	1.1B	1.1B
Avg. Success Rate	93.8%	92.0%	92.6%	93.3%

Table 5. INT4 quantization results for π_0 with LIBERO.

Task	#Trials	#Successes			
		FP	PTQ	QAT	SQIL
(a)	30	25	22	25	24
(b)	30	24	20	22	24
(c)	30	22	18	17	21
Suc.Rate(%)		79	67	71	77

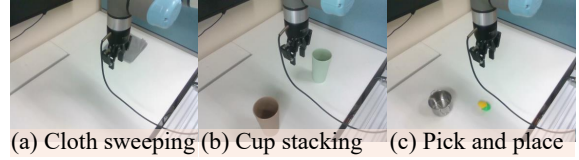


Figure 7. Real-world evaluation with *OpenVLA* on UR5.

ment occurred on the D4RL benchmark [16] using LSQ for both weights and activations at 4-bit precision.

5.2. Experimentals: Robot Manipulation

5.2.1. Full-Benchmark Evaluation

Table 3 outlines the comparative performance results on the LIBERO benchmark using various quantization methods. When PTQ (AWQ) applied, there is a notable decline of 3.2% in performance relative to the baseline, a critical issue for robot manipulation tasks due to their reliance on safe real-world interactions. Conversely, QAT achieves a modest performance increase of 0.7%, yet it still falls short of the baseline. Employing SQIL, which integrates additional insights from the FP policy on critical-state, significantly reduces the performance disparity to the FP model. This method shows a performance improvement of 2.3% over AWQ, underscoring its efficacy in resource-limited settings. Furthermore, we expand our assessment to include the state-of-the-art quantizer, QuaRot [12], with SQIL consistently matching the FP baseline success rate, thus confirming its general effectiveness.

5.2.2. Generalization

To assess the generalization capabilities of the quantized policy, we conduct a detailed evaluation as outlined in Table 4. We investigate the performance of the IN4 policy, quantized using SQIL, in responding to a variety of language instructions aimed at achieving a single goal. The

results show that the IN4 policy’s performance closely matches that of the FP policy. Additionally, we analyze how this policy performs under different lighting conditions, demonstrating its robustness by maintaining generalization performance at 80% and 60% brightness levels.

To further assess the generality of SQIL, we applied it to π_0 [2], a compact and high-performance policy obtained via flow matching-based action generation. As shown in Table 5, SQIL consistently outperforms PTQ and QAT, achieving performance close to the FP baseline while providing 3.0 $\times$ memory savings.

Fig. 7 compares the performance results of various quantization techniques in real-world scenarios. Relative to the FP policy, the performance of PTQ significantly declines, while QAT exhibits slight improvements. Notably, SQIL achieves performance comparable to the FP policy, demonstrating that reducing quantization errors in important states directly contributes to achieving goals in real-world environments. Additional results on the BridgeDataV2 [48] setup are provided in the *supplementary materials*.

Method	Bit-width	Suc. Rate % $\uparrow$				Dri. Score % $\uparrow$			
		tt	tn	nt	nn	tt	tn	nt	nn
Baseline	FP	82	74	80	68	78	71	80	64
PTQ	W4A4	34	43	36	29	30	38	35	31
QAT	W4A4	62	58	58	48	62	62	58	58
SQIL	W4A4	80	72	72	68	76	70	74	61

Table 6. Comparative analysis of success rate, driving score, and reward across various quantization methods using *CILRS* on the NoCrash-dense benchmark, differentiated by combinations of training and new conditions in town and weather (tt, tn, nt, nn).

TASK	FP32	INT4	
	IL	LPPD[36]	SQIL
Cartpole Balance	652	424	635
Walker Stand	692	442	688
Hopper Stand	645	404	643
Cheetah Run	567	294	556
Finger Spin	684	421	640
Humanoid Stand	565	356	590
Humanoid Walk	550	205	535

Table 7. Comparison of average return for each task by quantization method on physics simulation on MuJuCo.

5.3. Cross-Domain Results

End-to-End Autonomous Driving. As seen in Table 6, the performance comparison after applying quantization methods shows a significant drop in performance in both train and new-towns under the W4A4 setting. However, by applying SQIL, the alignment with the driving capabilities of the FP policy is significantly improved, achieving much better performance. An example illustrating the differences in driving behavior across quantization methods is provided as a video in the *supplementary materials*.

Physics Simulation Tasks. To evaluate the generalization capability of our algorithms, we expand our evaluation to classical continuous control tasks from the DeepMind Control Suite within D4RL [16], as shown in Table 7. This allowed us to compare our quantization techniques SQIL, against established methods like LPPD [36].

5.4. Implementation

We evaluate the versatility of our quantization techniques by implementing them on various resource-constrained platforms, including an edge device (NVIDIA Jetson AGX Orin) and a low-end GPU (NVIDIA 2080Ti). For practical deployment, we test the autonomous driving models: the W8A8 model on a CPU (ARM Cortex-A78AE) and the W4A4 model on the NVIDIA 2080Ti GPU. Additionally, 8-bit and 4-bit weight-quantized VLA models are implemented on an Jetson AGX Orin’s GPU. Latency is measured using 1,000 input samples, while CPU and GPU energy consumption are recorded with tegrastats [39]. Detailed experimental settings, latency breakdowns, and further analyses are provided in the *supplementary materials*.

Autonomous Driving: We employ ncnv [38] for W8A8

Measurement	CPU (Jetson AGX Orin)			GPU (RTX 2080Ti)		
	FP32	FP16	W8A8	FP32	W8A8	W4A4
Memory (MB)	78.4	39.2	19.6	78.4	39.2	19.6
Latency (ms)	212.2	176.4	124.7	11.6	4.1	3.1
SpeedUp	1.0 $\times$	1.2 $\times$	1.7 $\times$	1.0 $\times$	2.9 $\times$	3.7 $\times$
Energy Saving	1.0 $\times$	1.1 $\times$	1.3 $\times$	1.0 $\times$	2.5 $\times$	3.1 $\times$

Table 8. Performance Metrics for *CILRS* (14.5 GFLOPs) measured on a CPU and a GPU, demonstrating memory consumption, latency and energy saving across the different precision formats.

Weight Type	<i>OpenVLA</i> (> 2.0 TFLOPs)			
	Memory	Latency	SpeedUp	Energy Saving
BF16	15.2 GB	955.2 ms	1.0 $\times$	1.0 $\times$
INT8	7.9 GB	573.6 ms	1.6 $\times$	1.7 $\times$
INT4	4.0 GB	374.7 ms	2.5 $\times$	2.5 $\times$

Table 9. Performance Metrics for *OpenVLA* on GPU (NVIDIA Jetson AGX Orin): Memory, latency, and energy saving across various precision formats.

support with ARM NEON intrinsic and AutoTVM [5] for W8A8 and W4A4 GPU kernels. As shown in Table 8, the *CILRS* (CPU) W8A8 model achieves a 1.7 $\times$ reduction in latency compared to FP32, along with a 75% reduction in memory usage and a 30% improvement in energy efficiency. For the *CILRS* (GPU), the W4A4 model achieves up to 3.7 $\times$ lower latency and 3.1 $\times$ better energy efficiency.

Robot Manipulation: For GPU implementation of weight-only quantized *OpenVLA*, we use TensorRT for the vision encoder and MLC-LLM [45] for the LLM backbone. Compressed weights mitigate the memory-bound nature of LLM operations, significantly accelerating inference. As shown in Table 9, the INT8 model achieves a 1.6 $\times$ speedup, while the INT4 model achieves a 2.5 $\times$ speedup compared to FP. Energy efficiency improves by 1.7 $\times$ with INT8 and 2.5 $\times$ with INT4. Additionally, the 4 $\times$ memory reduction with INT4 is particularly advantageous for foundation models, which are inherently memory-intensive, making this approach suitable for edge devices with limited resources.

Our evaluations on edge devices demonstrate that the optimized models significantly improve performance and energy efficiency on both CPU and GPU platforms. These results validate that our quantization techniques enable resource-efficient robot control in computationally constrained environments.

6. Conclusion

This paper proposes a quantization framework for IL-based models that enhances robustness against low-bit precision errors, ensuring efficiency on resource-limited hardware. Evaluations on robot manipulation and self-driving models show superior speedups and energy savings on real CPU and GPU, closely preserving full-precision accuracy and demonstrating practical deployment potential.

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